

PROJECT OVERVIEW

HR Attrition Analysis

An end-to-end data project using SQL and Power BI to uncover why employees leave — and what to do about it.

THE PROBLEM

Employee attrition is costly and disruptive — yet most organisations lack a clear, data-driven view of who is leaving, why, and when. This project was built to answer those questions systematically using 1,470 employee records spanning demographics, job roles, satisfaction scores, and behavioral patterns.

TOOLS & TECHNOLOGIES

- SQL (MSSQL)** — Data cleaning, staging, aggregations, window functions
- Power BI** — DAX measures, Power Query (M), interactive dashboards
- Power Query** — Data type handling, custom column transformations

1,470

Employee Records Analysed

5

Analytical Dimensions Explored

16.12%

Overall Attrition Rate Identified

KEY FINDINGS — POWER BI VISUALISATION

OVERTIME

41.86%

Employees who work overtime and travel frequently face the highest attrition in the entire dataset — nearly 3× the company average. Overtime is the single strongest risk multiplier identified.

WORK-LIFE BALANCE

31.25%

Employees scoring their WLB as 'Poor' (Score 1) show 31.25% attrition — more than double those rating it 'Good'. WLB scores function as a leading indicator before formal resignation signals appear.

JOB ROLE

39.76%

Sales Representatives face the highest role-level attrition at 39.76%. Lab Technicians (23.94%) and HR employees (23.08%) also significantly exceed the 16.12% company benchmark.

JOB LEVEL (SENIORITY)

26.34%

Entry-level employees (Level 1) attrition rate is 26.34% — nearly 6× that of Level 4 (4.72%). This points to clear gaps in onboarding quality, early career support, and role clarity.

BUSINESS TRAVEL

24.91%

Frequent travelers show 24.91% attrition vs 8.0% for non-travelers. Travel frequency is one of the most controllable and policy-addressable drivers identified in the analysis.

HOW I BUILT IT IN POWER BI

Power Query

Grouped employees by demographic attributes — age bands, education levels, distance categories — using custom M code columns and conditional logic

DAX Measures

Built Attrition Rate %, Active Employee Count, and segment-level attrition measures using CALCULATE + FILTER for dynamic cross-filtering

Dashboard Pages

Designed 5 themed pages: Executive Summary, Travel & Overtime, Job Role & Level, Satisfaction & WLB, Distance & Demographics

Interactivity

Added department slicer, drill-through on job role, and conditional formatting on the Age × Education matrix to surface risk at a glance

RECOMMENDATIONS & PROJECT IMPACT

Actionable outcomes and the value this analysis delivers to the organisation

Target the highest-risk employee segments first

- Conduct structured exit interviews for all Sales Representative and Level 1 employee departures
- Cap mandatory overtime for Travel_Frequently employees — the 41.86% attrition rate demands immediate policy intervention
- Assign onboarding mentors to all new Level 1 hires within the first 90 days of joining

Reduce structural overtime and travel burden

- Review workload distribution across Sales and HR — both departments show overtime overlap with high travel frequency
- Introduce compensatory leave policies for frequent travelers to reduce cumulative burnout risk
- Benchmark Sales Representative compensation against market — pay misalignment is a common driver at this attrition level

Embed real-time satisfaction monitoring in HR

- Deploy quarterly pulse surveys tracking WLB, Job Satisfaction, and Environment Satisfaction scores by team
- Set automated HR alerts when team-level satisfaction scores fall below Score 2 — act before resignations occur
- Link people health metrics to manager performance KPIs to create leadership accountability for retention

Build a predictive attrition early-warning model

- Model employee risk scores by combining travel frequency, overtime, satisfaction scores, and distance from home
- Flag high-performance, low-satisfaction employees as priority retention targets before they disengage
- Design role-specific career progression tracks for HR staff to reduce mid-seniority attrition